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FOR ENTREPRENEURS, CHOICE MATTERS

Multiple Paths to Value: Test Two, Choose One

By JOSHUA GANS and SCOTT STERN

In the late '90s, when the internet was still in its infancy, several U.S. entrepreneurs hatched almost the exact same business idea around the same time: a platform to buy groceries online and have them delivered to your home. They all planned to launch in big cities, where delivery was cheaper, and they all promised convenience, time savings and user friendliness for busy professionals. They diverged on one key aspect, though. Was it better to create an online store that substituted for traditional grocers? Or was it better to collaborate with existing stores and serve as their online sales channel?

This situation – where multiple startups come up with the same basic idea yet stake their futures on sharply different visions of how to realize it – goes with the territory of entrepreneurship. Realistically, there are only so many large-scale experiments that can be carried out to test which is the better way to go.

In the case of online grocery shopping, two bets were placed: Webvan developed its own vertically integrated store, while Peapod

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outsourced its online sales service to existing grocers. In the end, neither set the world on fire – though Peapod has managed to survive, unlike Webvan, which burned through millions of dollars of investor funds before tanking.

These are familiar tensions: there are multiple pathways to create and capture value, but limited resources will constrain you to choose just one; that choice is crucial, but working out which path to take requires knowledge that can only be gleaned through experimentation and learning; yet the process of experimentation and learning inevitably requires some level of commitment that forecloses certain options. Webvan cannot be blamed for not knowing in advance which strategy would work, just as it could not afford to experiment with an endless number of business models until it found the right one. And once it had chosen to go big and try to disrupt supermarkets, it could not easily turn around and cooperate with those supermarkets if it all failed. This is the paradox of entrepreneurship.

So, what should entrepreneurs do? In our years of research and through experience with startups both at MIT and the University of Toronto, we have found certain strategies that can be adopted to hone and optimize such choices. This article outlines three principles to help business strategists beat the paradox of entrepreneurship, whittling down their choice sets until they are faced with two equally viable alternatives. That is when to "test two, choose one," which is the recommended approach for decision-making under conditions of uncertainty.

EXECUTIVE SUMMARY

You have an idea for a new business, but there are multiple paths to create and capture value. To choose a path, you have to experiment, but experiments are costly and require at least a partial commitment, which consequently alters the nature of the path, precludes other paths, or worse, takes you down a path of no return. This article describes this entrepreneurial paradox and summarizes the key factors that need to be weighed at every step to hone and optimize make-or-break choices. Although the strategic considerations presented in this article are common to any competitive strategy, the authors discuss the specific ways in which entrepreneurial strategy is different. Rather than using experiments to justify a particular choice, the goal is to whittle down choice sets until you are faced with two equally viable alternatives. That is when to "test two, choose one," which is the recommended approach for decision-making under conditions of uncertainty.

The Dilemmas of Entrepreneurial Strategy

Entrepreneurial strategy is much like growing a large decision tree. Starting with the seed of an idea, the founders nurture and then prune choices in order to transform that idea into an organization that eventually flourishes. Transitory by nature, entrepreneurial strategy is not a fixed document or business plan. Rather, it is a set of choices that the founders and their teams must make to test various hypotheses related to value creation and value capture. It entails experimentation and a partial commitment, at the very least.

When it comes to testing hypotheses, most businesses are spoiled for choice. Take Dropbox. As an MIT student, Drew Houston identified a common problem: repeatedly forgetting your USB but still needing access to your files. There were various possible ways to address this. He could have developed an in-house enterprise solution. He could have partnered with software companies to develop an app. Or he could opt to synchronize files across computers using cloud computing, which, as we now know, is how he eventually developed a multibillion-dollar business. But Houston's initial dilemma serves to highlight several crucial axioms.

AXIOM 1: THERE IS MORE THAN ONE ALTERNATIVE PATH BY WHICH VALUE CAN BE CREATED AND CAPTURED FOR A GIVEN IDEA. Startups have arguably the broadest set of alternatives for creating and capturing value from an idea and the greatest

> Entrepreneurs cannot know in advance which business model will work, nor can they experiment with them all. They have to make choices, and their choices will be consequential.

freedom of choice. However, as stated in the next axiom, they face severe limitations when it comes to exploring those alternatives.

AXIOM 2: RESOURCE CONSTRAINTS PREVENT PURSUIT OF MORE THAN ONE ALTERNATIVE AT A TIME. Most startups do not have sufficient resources or strategic flexibility to explore multiple commercialization paths for a given idea. Instead, they must balance the tension of having an extensive choice of paths but not having enough resources to pursue them all at once.

AXIOM 3: UNCERTAINTY PREVENTS A RANKING OF PATHS. Entrepreneurs usually have so little ex-ante information that attempting to rank the alternatives is impractical. That is why early-stage entrepreneurs need to engage in commitment-free experimentation, to develop an understanding of the relative merits of various approaches.

AXIOM 4: LEARNING REQUIRES MAKING A PARTIAL COMMITMENT; BUT COMMITTING YOURSELF TO ONE PATH CONDITIONS THE OTHERS. There is no getting around the fact that, to test the viability of an idea, you have to actively experiment. But by experimenting, you start to shape the market and influence other paths, even to the point of making those others unviable.

In summary, entrepreneurs cannot know in advance exactly which business model will work, nor can they experiment with them all. They have to make choices without enough information to rank those choices, and their choices will be consequential. Under such circumstances, the key is not to commit yourself until you reach a point where you have two equally viable and attractive alternatives that cannot be ranked without making some commitment to one of them. To guide entrepreneurs through this complex process, we elaborate on three basic principles that can help.

PRINCIPLE 1: CHOICE MATTERS

The first principle that entrepreneurs need to grasp is that their choices matter. The reason they matter is precisely because of the path dependence mentioned earlier. Pursuing a certain strategy, even if only through tentative experiments, is effectively narrowing your choice set, precluding some paths or taking you down a path of no return.

Entrepreneurial choice needs to be framed as an action that (a) triggers experimentation and (b) tests a specific element of the value-creation-and-capture hypotheses. The testing of *a specific element* is key. How to determine that?

Again, it would be inefficient, if not counterproductive, to try to establish conclusive links between a change in each and every variable and outcomes. Given the multitude of variables – owing to unresolved uncertainty among multiple alternatives along multiple dimensions – it is simply not feasible to carry out many experiments. You have to prioritize, and the best way to do that is to test only those variables that are the decisive determinants of your business identity and mission. That brings us to the next principle.

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PRINCIPLE 2: THESE CHOICES MATTER

The second principle highlights the four most salient choices that startups face. These choices relate to customers, technology, identity and competition. The first two – customer segmentation, and selecting or creating a technology that facilitates value delivery – are vital for creating value. The other two – establishing the identity the venture will inhabit, and articulating with whom and how it will compete – are vital for capturing it.

CHOOSING A CUSTOMER. Business schools often teach frameworks whereby you take the value that different customers create, you subtract the costs incurred for serving each type of customer, and then you rank and select the customer segment that maximizes value creation. This may work for established companies with existing customer bases. But for startups with no track record, it is not so easy to figure out which customers will yield what value, and how much it will cost.

Fortunately, a great deal of research exists on how consumers adopt new ideas and technologies, and on innovation diffusion – the process by which new ideas and technologies spread among customers and markets. In general, diffusion is slow. Those ideas that do make it tend to follow an S-shaped pattern as they move across different adopter categories, from early adopters until finally reaching the mass market. According to organizational theorist Geoffrey Moore, the big challenge is gaining traction with the early customer segment and then "crossing the chasm" to serving more mainstream markets. This implies that entrepreneurs should choose an initial segment of customers who are influential on other segments.

Entrepreneurial strategy differs slightly in that, rather than seeking out the ideal early adopter or beachhead segment, the choice of customer is often between numerous equally viable segments and depends on other factors. If your tech startup, for example, aims for a broad standard, it makes sense to prioritize mainstream customer segments at an earlier stage. If, on the other hand, you plan to be at the bleeding edge of innovation, prioritizing customer segments that are tolerant of failure may be a better choice. Whichever initial segment you select will determine which of the many potential market S-curves your startup will travel along.

CHOOSING A TECHNOLOGY. All technologies have the potential to create significant value. To tap into that value, entrepreneurs need to weigh the trade-offs between the benefits of investing in exploration through research and experimentation, on the one hand, and the gains from exploitation through commercialization, on the other.

Borrowing loosely from James March's work on the trade-off between allocating resources in organizational learning, we have plotted a graph with the level of technology on the Y axis and time on the X axis (see **Exhibit 1**). The slope represents the pace at which the technology can be expected to improve on a particular performance metric over a given period of time. This rate follows an S-shaped curve, reflecting a period of technology exploration, followed by a period of technology exploitation.

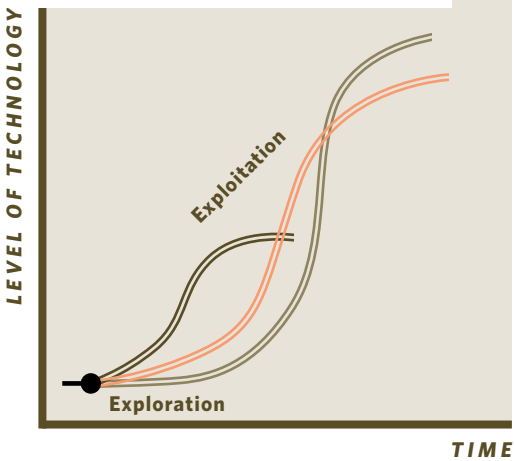
The choice of exploration vs. exploitation induces alternative S-curves. The job of the founding team is to select from among these alternative paths by weighing up the pros and cons of adopting an established technology of known quality vs. an emerging technology of unknown potential.

Carefully considering the myriad implications of technology adoption and, if applicable, technology innovation, and how they cohere with your firm's choice of customer, competition and identity, can help you define a clearer

EXHIBIT 1

Weighing the Tech Trade-offs

THE CHOICE OF EXPLORATION VS. EXPLOITATION INDUCES ALTERNATIVE S-CURVES.



> You should view the ecosystem you participate in as a choice, and consider how different geographic and network locations might impact other elements of your firm identity.

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choice of technology as well as think through your overall entrepreneurial strategy.

For example, if you are planning to cooperate with established firms through licensing or building a key part of their value chain, you need to consider how to integrate into existing systems without creating prohibitive friction for incumbents.

On the flip side, if you are planning to compete with established firms, you must be able to find a way of establishing a competitive advantage with this choice of technology before well-resourced incumbents erode it.

CHOOSING AN IDENTITY. When it comes to molding the qualities, traits and characteristics that will define a nascent firm's identity, founders face a unique dilemma. Without an extant track record, they have no core competencies, firm culture or reference firms in analogous industries to anchor (or constrain) their vision for the future of the firm. What they have, instead, is a substantial depository of prior knowledge and well-established networks. It is up to the founder(s) to choose which elements of those knowledge and network bases to leverage for the entrepreneurial endeavor.

In other words, rather than assuming that your firm has a fixed, predestined identity or DNA based on its founders' prior knowledge, think about identity as a choice between different DNAs of the firm. Each potential strand of DNA is a unique combination of distinct elements of identity drawn from the following:

- **Founder Purpose.** When it comes to determining the purpose, mission and definition of success for their startup, founders have a great deal more latitude in choice than is usual. For example, Anita Roddick founded The Body Shop with no training or experience and a simple goal of making a living. Early on, she spotted an opportunity to orient the business around a social cause she supported: advocating for ethical, sustainable cosmetics. This choice of purpose shaped its subsequent product offerings, capabilities and positioning, leading the company to become a leader of a whole new category of ethical cosmetics.

- **Internal Capabilities.** The choice of early hires, the organizational design (including incentives and team responsibilities) and the early capital investments will mold internal capabilities and collectively shape the firm's culture.

- **External Positioning.** The firm's positioning and reputation are other important facets of identity. Elon Musk could have positioned Tesla as an innovative entrant in the general market for electric vehicles, choosing to gain market leadership that way. Instead, he chose to enter the high-end luxury market. Not only did this market have many more incumbent players to compete with, but this positioning forced Tesla to be on the leading edge in many areas, in terms of car size, comfort, material quality and connected apps.

- **Broader Ecosystem.** Choosing to set up in a particular location opens an entrepreneur to local peer efforts and knowledge spillovers. It also begins imprinting the internal and external identity of the firm. As such, you should view the ecosystem you participate in as a choice, and consider how different geographic and network locations might impact other elements of your firm identity.

CHOOSING THE COMPETITION. Most ventures are founded on one idea, based around one of three competitive propositions: a consumer insight, such as Facebook's intuition that college students could benefit from an online social network; a technology breakthrough, such as Google's development of a more effective algorithm for internet searches; or an idea for building a different type of company, such as Amazon's uncompromising cost-optimization model.

Based on its competitive proposition, the key choices for most startups are deciding with whom and how to compete.

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With whom to compete: competition vs. collaboration? Founders must decide whether or not their idea's route to commercialization lies in cooperating with established players or constructing a wholly independent value chain, which in most cases will compete with established players.

More specifically, they can choose to compete in the product market directly with incumbents, or they can operate in the ideas market by collaborating with established firms through licensing, alliances and/or acquisitions. Both have benefits and drawbacks.

By constructing its own value chain, a startup can avoid granting control rights over the ways in which their idea is commercialized to incumbent players, who might have incentives to thwart or slow down the entry of a startup's innovation if it threatens current offerings. They also escape the inevitable hassles and risk of expropriation that come from working with established firms. This results in greater operational freedom and the ability to capture value once the value chain is in place.

This is not to say that going it alone is easy. For smaller startups, establishing their own value chain from the bottom up is often prohibitively expensive. A more cooperative approach with established players can reduce the costs of market entry for a startup. It can also enhance the scope and speed of diffusion of the idea, and lessen the threat of competition from large incumbents.

If opting for a more cooperative approach, you need to focus on aspects that reinforce value for the customers of the established firms with whom you choose to collaborate. You also need to find ways to integrate your technology choices with those firms' systems and offerings. Whereas a company that chooses to compete with incumbent firms must develop the full value chain, a more cooperative approach requires a team that can work in an effective way with larger organizations.

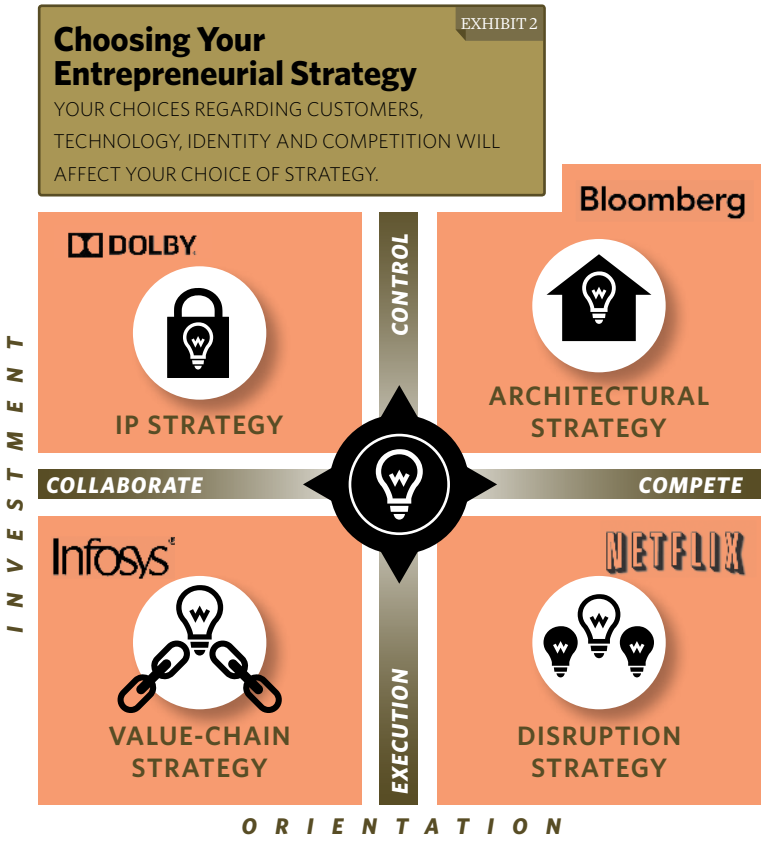
Ultimately, the choice between competition and cooperation is fundamental and drastically shapes the direction of the venture. Even though entrepreneurs cannot choose not to compete, if they decide to collaborate with established firms, they will face a more limited but less dangerous competitive landscape than if they opt for a more combative approach.

How to compete: execution vs. control? In deciding how to compete, there are two main options.

Control involves taking measures to prevent the idea from being exploited by others, such as protecting intellectual property. Control is about using knowledge and expertise to build entry barriers that can protect the firm against competition.

Conversely, execution means moving rapidly to commercialization and development, which involves more partners. In this case, a venture would want to leave entry barriers low, expecting and even perhaps welcoming competition.

From a customer choice perspective, a control-oriented startup may position itself to broader markets, whereas founders who focus on execution are more likely to choose a narrower customer segment. Execution-oriented startups prioritize learning from early adopters and are more likely to iterate. They ride the S-curve, rather than focusing on transferable, generalizable technologies that are subject to standardization.



PRINCIPLE 3: THESE CHOICES MATTER TOGETHER

The third principle discusses what the previous choices imply for strategic decision-making. For any given idea, the following alternative

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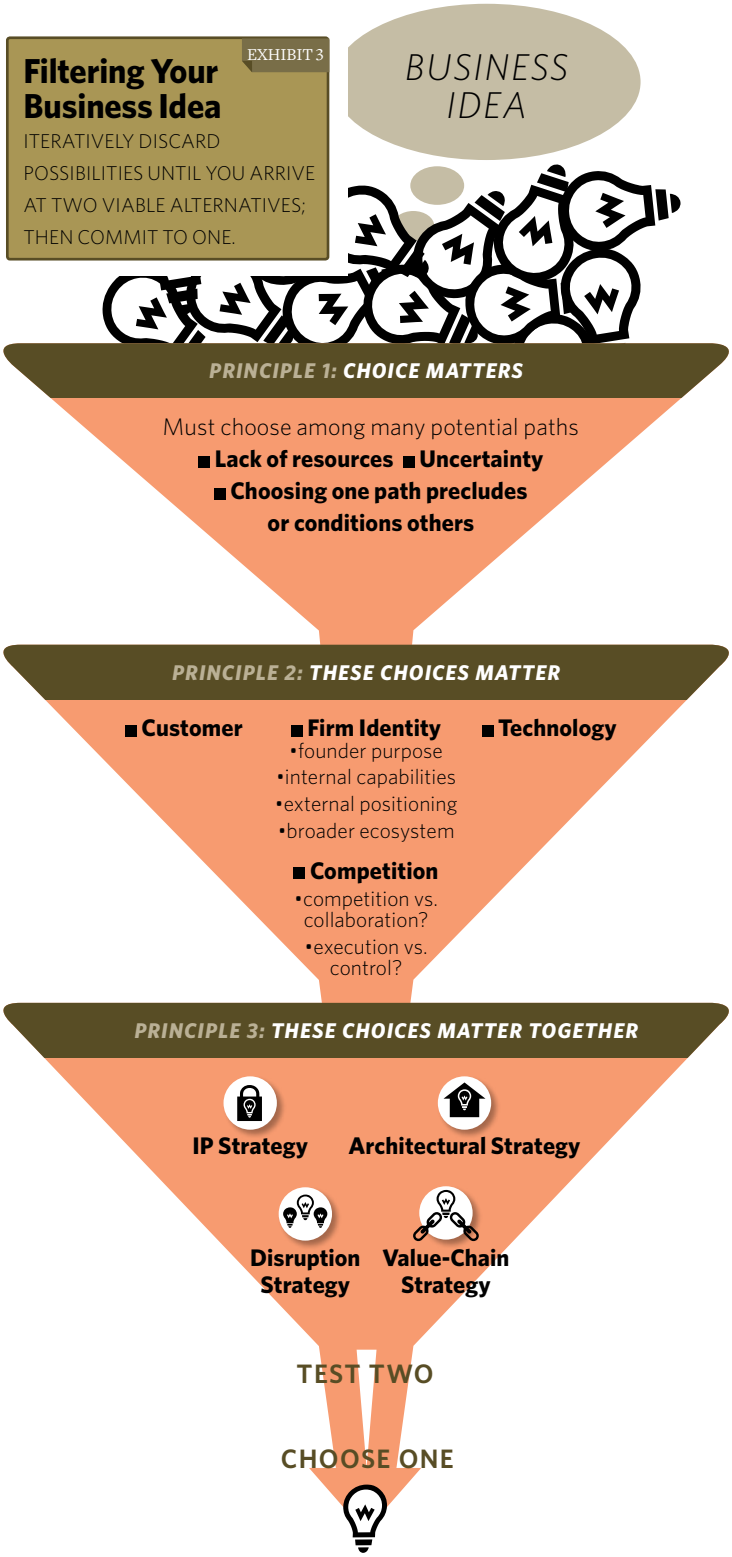

trajectories need to be contemplated for organizing entrepreneurial strategy. Each of these strategies involves a distinct set of choices regarding customers, technology, identity and competition (see **Exhibit 2**).

IP STRATEGY. Entrepreneurs pursuing an IP strategy choose to invest in control over their underlying innovation. They create value by working in collaboration with incumbent firms in an industry. By this means, they develop transferable innovations that create value for end customers.

Dolby exemplifies this strategy. Despite having limited direct interaction with film directors, music producers or audiophiles, Dolby's patented sound technologies have become a global standard in audio for music and films. Dolby achieved this by licensing its proprietary technology to audio product developers and manufacturers, including Sony, Bose, Apple and Yamaha. By serving end customers in an arm's-length way, companies that follow an IP strategy, like Dolby did, can obtain extraordinary returns.

DISRUPTION STRATEGY. A disruption strategy involves prioritizing execution and flexibility over control. Companies that pursue this strategy choose to engage industry incumbents in direct competition in the hope of rendering their business models obsolete.

A classic example of this is Netflix. Born out of the widespread frustration with traditional video-rental overdue fines, the founders envisaged a video-rental delivery service that would leverage the then-emergent technology of DVDs. After testing the concept by mailing a DVD through the U.S. postal system, Netflix carved out a novel video rental niche that allowed film buffs to rent online, and receive and return DVDs through the mail. Netflix has continued to invest in execution, evolving from sending DVDs through the mail to streaming content direct to homes via the internet, bypassing or riding over the top of traditional gatekeepers. Now, it is producing its own original content, once again going head to head with established players. Given how Netflix supplanted its main rival Blockbuster, traditional content producers may be justifiably worried.

VALUE-CHAIN STRATEGY. Entrepreneurs who pursue a value-chain strategy also choose to invest in execution rather than control; the main difference is they create value through cooperation with incumbents.

An example here is the Indian tech services provider Infosys. By leveraging IT and software developments, Infosys helped clients such as Accenture, Ernst & Young and Deloitte tap into a skilled global workforce, particularly in countries like India. It embraced collaboration with those firms by positioning itself as a key

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broker in an established value chain, and then invested in execution to become an integral part of that chain.

ARCHITECTURAL STRATEGY. As with an IP strategy, those who pursue an architectural strategy also invest in control over their underlying innovation; the distinction here is that entrepreneurs aim to compete with incumbents by developing a whole new value chain in the industry.

Michael Bloomberg did this when he launched his trademark computer terminal to provide analysts with real-time financial data. Besides product innovation, Bloomberg engineered a unique business model in which Bloomberg retained actual ownership of the terminals, selling subscriptions on a per-user basis. Over time, Bloomberg leveraged its desktop presence to expand into secure messaging services, as well as broader financial news aggregation, which made imitation all the more costly. Over the past three decades, Bloomberg has secured and retained a market leadership position, with over 300,000 subscribers for its terminals, translating to billions of dollars annually.

Companies that follow an architectural strategy must negotiate a terrain in which entrepreneurial strategy and traditional strategy overlap: entrepreneurs invest in control over their core idea while competing with established firms. Unlike the other strategies, this one is about "architecting" an entirely new value chain in the marketplace, rather than boosting the performance of an existing chain or trying to render the other ones obsolete.

Test Two, Choose One

The main goal of going through all the previous choice sets is to iteratively discard strategic alternatives using commitment-free analyses until you reach the paradox of entrepreneurship (see **Exhibit 3**). As stated before, this is the point where you are faced with at least two equally viable and attractive alternatives that cannot be ranked without making some level of commitment to one of them. This is when you need to carefully develop two fully fleshed-out models before choosing which of the alternatives to commit to.

The differences between these two models should not be trivial: a slight change in the target customer or a modest tweak to the supply chain will not suffice.

Instead, you need to develop a clear and compelling vision for your business that brings together all four core choices. Then, work through the logic of an alternative strategic trajectory, reconsidering each of the four choices.

What makes entrepreneurial strategy unique is that it challenges the notion of an "optimal" customer, technology, competitive market or founding team identity. By emphasizing the options that entrepreneurs have available – multiple, complementary combinations of the four core choices – entrepreneurial strategy encourages founders to make active choices concerning what capabilities to build, how to position their venture within a value chain and in the marketplace, as well as helping them explore and come to terms with how their values as founders shape the company they build.

Although the tool of "test two, choose one" sounds straightforward enough, it is by no means a quick, cheap or easy formula. Experiments can be expensive. Strategy formulation and implementation take time. The types of strategic commitments inherent in establishing competitive advantage often involve significant capital expenditures.

The alternative, however, is potentially a lot more costly. Making big choices up front, without properly considering the alternatives, locks ventures into commitments and activities that can be difficult, if not impossible, to unwind.

As Webvan learned the hard way, once you are fully committed, there is often no turning back. The choice-oriented sequence described in this article should help you place your bets more wisely. □

TO KNOW MORE

- The origins of this article come from past research conducted with co-authors David Hsu, Matt Marx, Fiona Murray and Kenny Ching, along with courses taught by Scott Stern at MIT and Joshua Gans at the University of Toronto. For more on the ideas in this article, see the paper "The Foundations of Entrepreneurial Strategy" co-authored with Jane Wu (MIT). Stern presented on this subject at IESE Business School as part of a Distinguished Research Seminar.